

Noah James

Bexley, Ohio, 43209 | noahcj500@gmail.com | (614) 483-6123

Education

Bachelor of Science, Software Engineering

Rose-Hulman Institute of Technology, Terre Haute, IN

Honors: Dean's List (Spring 2025, Fall 2025, Winter 2025-2026, Spring 2026)

Relevant Courses: Database Systems, Operating Systems, Intro to Web Programming, Software Quality Assurance, Computer Architecture I, Data Structures & Algorithm Analysis, Object-Oriented Software Development

May 2028

GPA: 3.35

Certifications and Honors

AP Scholar with Distinction: Awarded for outstanding academic performance on multiple Advanced Placement exams 2023–2024

Global Seal of Biliteracy – Working Fluency (Spanish): Awarded for high-level proficiency in all language domains

OHSBC Academic All-Ohio (Varsity Baseball): Awarded for academic excellence while competing at the varsity level (Bexley HS)

"Our Kinda Guy" Award: Selected by coaching staff for team leadership and character

Skills

Languages: C, CSS, HTML, Java, JavaScript, Python, SQL, TypeScript

Software: Git, React, Visual Studio Code, Eclipse, Linux, OnShape, AutoCAD, Google Workspace, Test-Driven Development

Project Experience

TopDown Tactix - <https://topdown-tactix.vercel.app/>

June 2026-Present

- Engineered a real-time, 2D multiplayer soccer game using a monorepo architecture with React (frontend) and Node.js/Socket.io (backend) to achieve precise client-server state synchronization.
- Developed a high-performance, deterministic physics engine with Matter.js, utilizing custom collision filtering to enforce specific interaction rules between players, boundaries, and the ball.
- Implemented sub-stepping integration and velocity clamping within the physics engine to ensure movement stability and prevent tunneling during high-velocity collisions.
- Established a comprehensive testing suite using Vitest to validate critical game systems, including goal detection logic, state resets, and server-side socket communication.
- Deployed automated CI/CD pipelines via Vercel and Render for production-ready hosting.

Forbidden Island

March 2026-May 2026

- Engineered a desktop implementation of the board game Forbidden Island in Java, managing complex game logic through polymorphic role abilities and dynamic state transitions
- Executed a rigorous Test-Driven Development (TDD) workflow, utilizing mutation testing and static analysis to achieve 100% basis path coverage on core business logic
- Integrated a ResourceBundle-based localization system to support dynamic language switching without modifying core game mechanics or Swing UI components.

RoseShreddedNerds Workout Tracker

December 2025-February 2026

- Developed a full-stack fitness tracking application for Rose-Hulman students using React and Python, enabling users to log workouts, track personal records, and join fitness classes
- Engineered a custom RESTful API and HTTP server using Flask and pyodbc to manage secure user authentication, session scheduling, and real-time exercise volume tracking
- Architected a relational database in SQL Server with stored procedures to automate complex fitness calculations, including Estimated 1 Rep Max via the Epley Formula and campus-wide leaderboards
- Designed a responsive frontend with Bootstrap and React Router, featuring a dynamic dashboard that visualizes weekly consistency streaks and multi-set workout history

Portfolio Website - <https://noahcjames.github.io/>

September 2024

- Designed and deployed a responsive portfolio website using HTML/CSS/JS and GitHub Pages, utilizing Git for version control and site maintenance

Fall Jump Video Game - <https://n04h500.itch.io/jump-fall>**September 2023-December 2023**

- Independently designed and developed a full video game from scratch using GameMaker Studio and GameMaker Language (GML)
- Learned and applied a new programming language while implementing core gameplay mechanics, debugging issues, and refining user experience
- Programmed player behavior and game logic for a complete playable level and published the game online for public access

Work Experience

Athletic/Event Staff, *Rose-Hulman SRC* - Terre Haute, Indiana**September 2025-Present**

- Set up and tore down multiple athletic and campus events, ensuring venues stayed on schedule
- Hosted visiting teams and officials, providing clear communication and hospitality throughout events
- Executed on-site responsibilities reliably to ensure athletic and campus events ran efficiently

Team Member, *Graeter's Ice Cream* - Bexley, Ohio**June 2022-September 2025**

- Collaborated with team members to deliver efficient, accurate customer service
- Accurately processed cash and credit transactions in a fast-paced environment
- Followed standardized procedures to maintain clean, organized operations and reliable store closing

Leadership Experience

Public Relations and Activities Chair, *Interfraternity Council***March 2025-Present**

- Directed IFC marketing, event advertising, and social media presence.
- Coordinated IFC philanthropy and campus events in collaboration with the Office of Student Affairs
- Served as the public voice of the Interfraternity Council, fostering inclusion and campus engagement

Activities

Rose-Hulman Institute of Technology

- Triangle Fraternity, *Rose Tech Chapter*; Career Services and Employer Relations Volunteer; Soccer Club